

Heuristic Evaluation

Expert Reviews

- More formal expert reviews have proven effective
- Variety of expert review methods:
 - **Heuristic evaluation**
 - Cognitive walkthrough
 - Guidelines review
 - Consistency inspection
 - Formal usability inspection

Heuristics

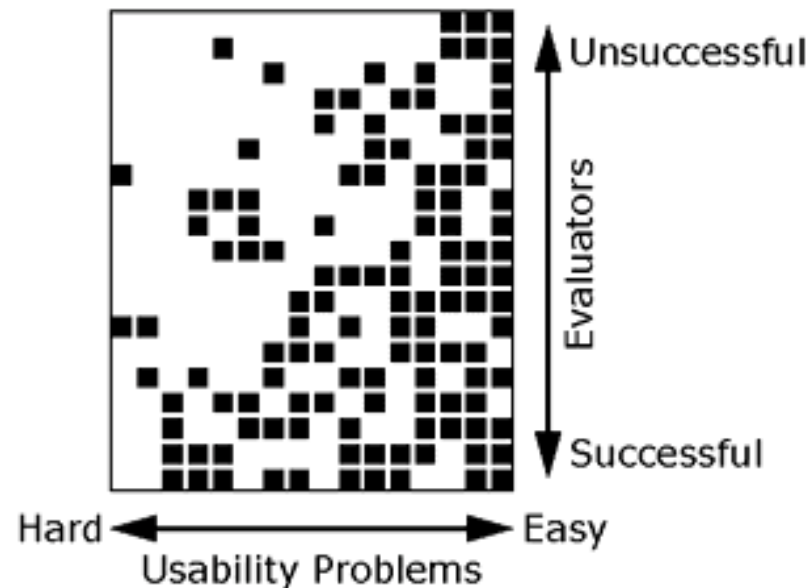
- Heuristics are collection of rules and guidelines
- Help to identify design problems
- Traditionally quite large collections
 - Brown 1988: 302 rules
 - Smith & Mosier 1986: 944 rules
- Nowadays lighter methods have become used
 - [Nielsen](#): 10 rules
 - [Schneiderman](#): 8 rules



Jakob Nielsen

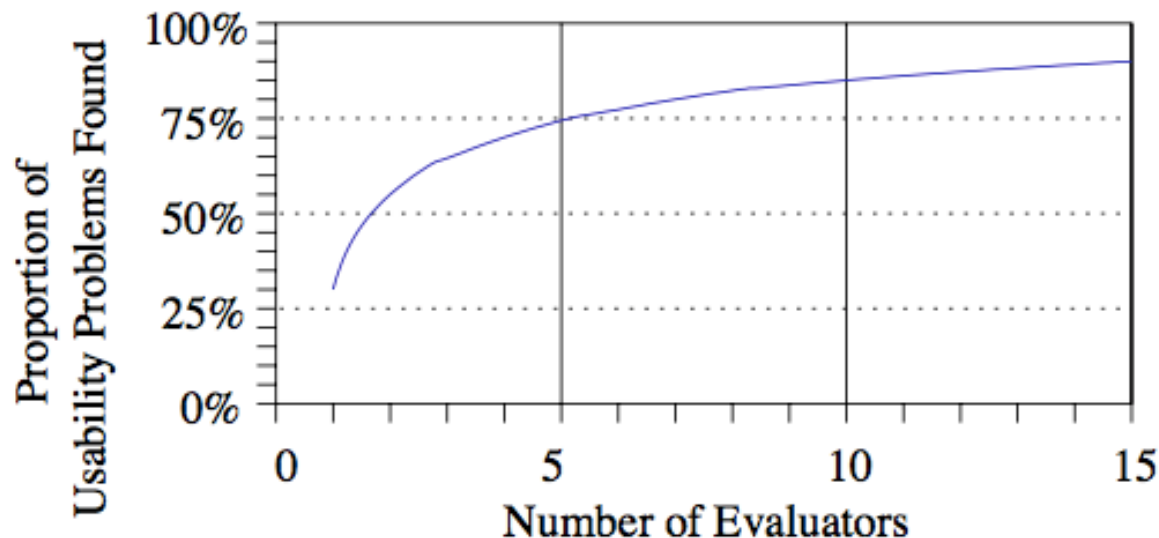
Heuristic Evaluation

- Different people identify different errors
- 19 evaluators were used to find 16 usability problems in a voice response system allowing customers access to their bank accounts (Image below - Nielsen 1992)
- Some usability problems are so easy to find that they are found by almost everybody
- Some problems that are found by very few evaluators



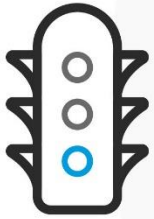
Expert Reviews

- Single evaluators found only 35 percent of the usability problems
- 3-5 expert reviewers can be highly productive!



- Cons:
 - Experts may not have an adequate understanding of the user groups

10 Usability Heuristics



**Visibility of
System Status**



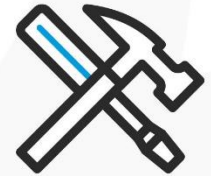
**Match Between System
& the Real World**



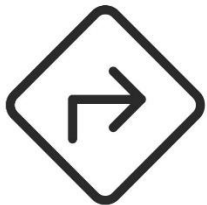
**User Control
& Freedom**



Consistency & Standards



Error Prevention



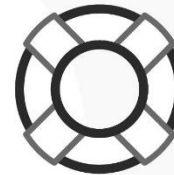
**Recognition Rather
than Recall**



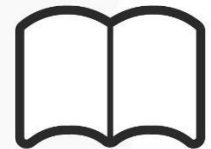
**Flexibility &
Efficiency of Use**



**Aesthetic &
Minimalist Design**



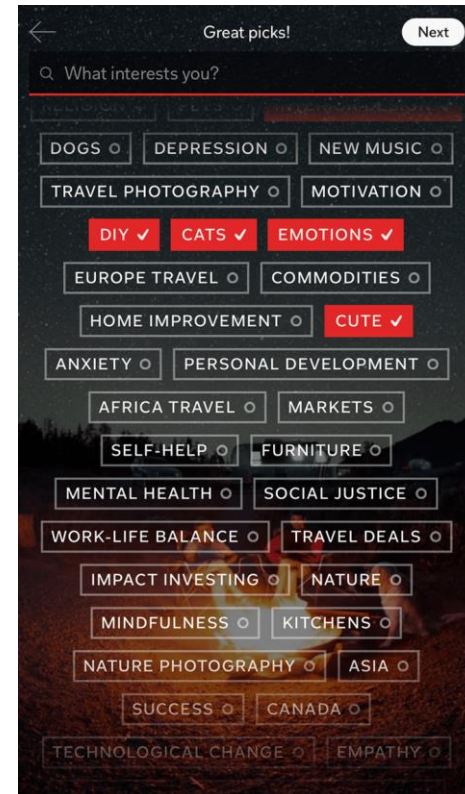
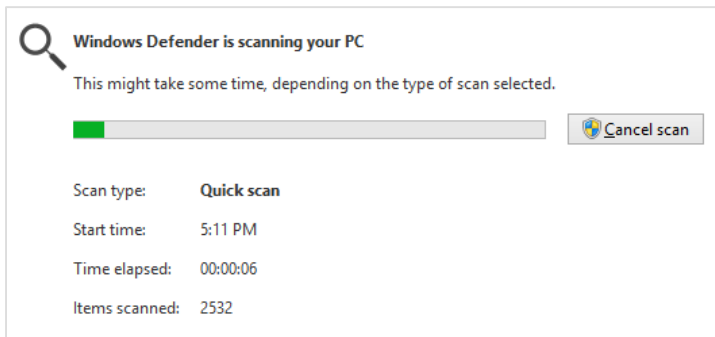
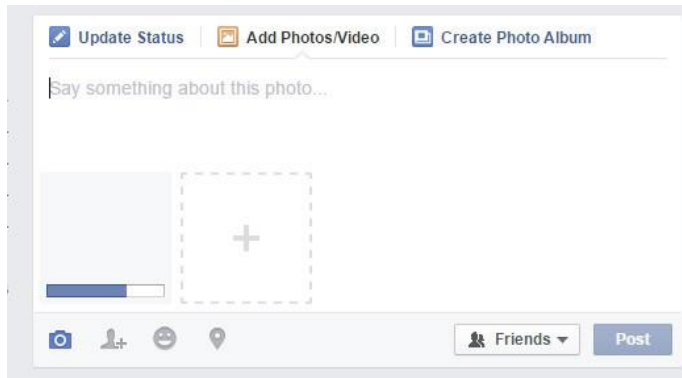
**Help Users Recognize, Diagnose
& Recover from Errors**



**Help &
Documentation**

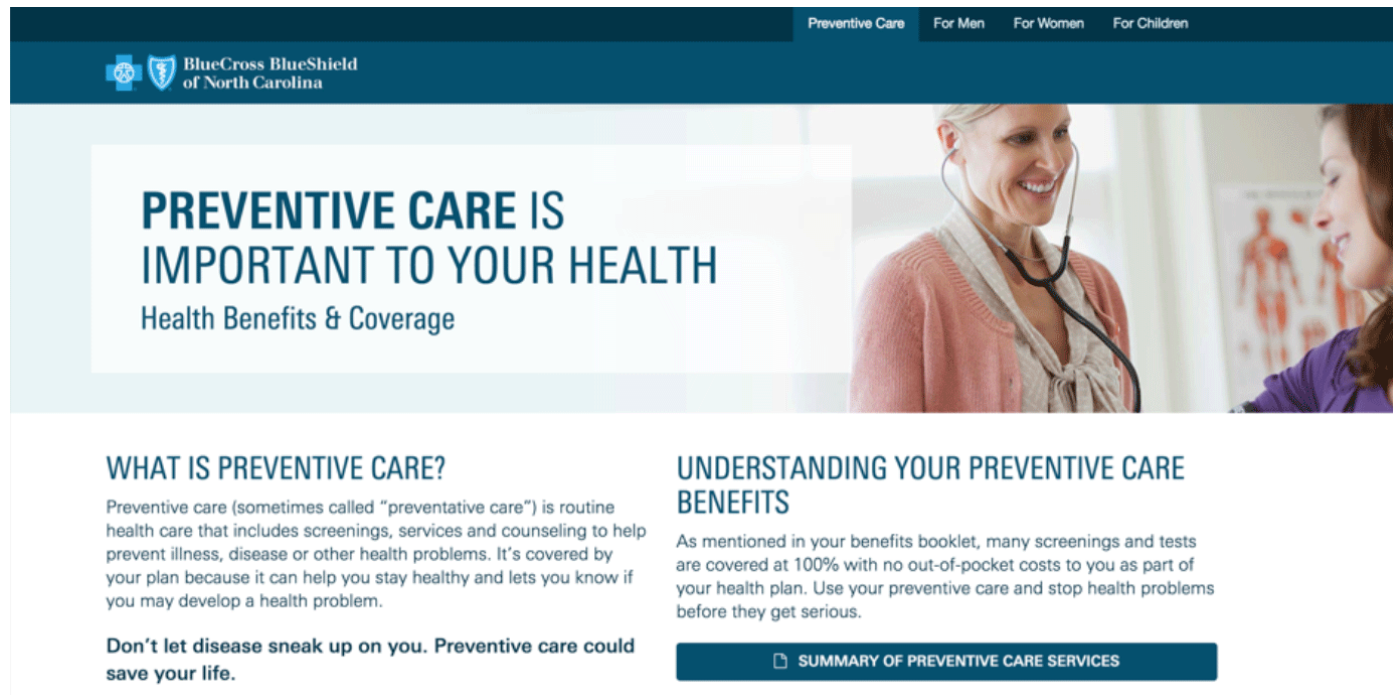
Visibility of System Status

- “A system should always keep users informed about what is going on, through appropriate feedback with reasonable time.”
- How well the state of the system is conveyed to its users



Match Between System and the Real World

- “The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.”
- BlueCross BlueShield of North Carolina uses clear headlines and relatable copy that’s easy to understand.



The screenshot shows the BlueCross BlueShield of North Carolina website. The header is dark blue with the logo on the left and navigation links for 'Preventive Care', 'For Men', 'For Women', and 'For Children' on the right. The main content area features a large headline 'PREVENTIVE CARE IS IMPORTANT TO YOUR HEALTH' with the subtext 'Health Benefits & Coverage'. To the right of the text is a photograph of a smiling female doctor with a stethoscope around her neck, talking to a patient. Below the headline, there are two columns of text. The left column is titled 'WHAT IS PREVENTIVE CARE?' and explains that preventive care includes screenings, services, and counseling to help prevent illness. The right column is titled 'UNDERSTANDING YOUR PREVENTIVE CARE BENEFITS' and states that many screenings and tests are covered at 100% with no out-of-pocket costs. At the bottom left, there is a bold statement: 'Don't let disease sneak up on you. Preventive care could save your life.' At the bottom right, there is a dark blue button with a white icon and the text 'SUMMARY OF PREVENTIVE CARE SERVICES'.

BlueCross BlueShield of North Carolina

Preventive Care For Men For Women For Children

PREVENTIVE CARE IS IMPORTANT TO YOUR HEALTH

Health Benefits & Coverage

WHAT IS PREVENTIVE CARE?

Preventive care (sometimes called “preventative care”) is routine health care that includes screenings, services and counseling to help prevent illness, disease or other health problems. It’s covered by your plan because it can help you stay healthy and lets you know if you may develop a health problem.

Don't let disease sneak up on you. Preventive care could save your life.

UNDERSTANDING YOUR PREVENTIVE CARE BENEFITS

As mentioned in your benefits booklet, many screenings and tests are covered at 100% with no out-of-pocket costs to you as part of your health plan. Use your preventive care and stop health problems before they get serious.

[SUMMARY OF PREVENTIVE CARE SERVICES](#)

Match Between System and the Real World

- In contrast, the About page for Abacus legal software's website contains overly technical language and jargon

The screenshot displays the AbacusNEXT website's navigation bar and a section titled "Abacus Private Cloud Quick Facts". The navigation bar includes the AbacusNEXT logo, a search bar, and links for "Contact Us" and "Get a Free Demo". Below the navigation bar, there are several menu items: "Private Cloud", "Business Software", "Accounting Solutions", "Legal Solutions", "Success Stories", "Partners", and "More".

The "Abacus Private Cloud Quick Facts" section features a grid of nine items, each with a green checkmark icon and a title. The items are:

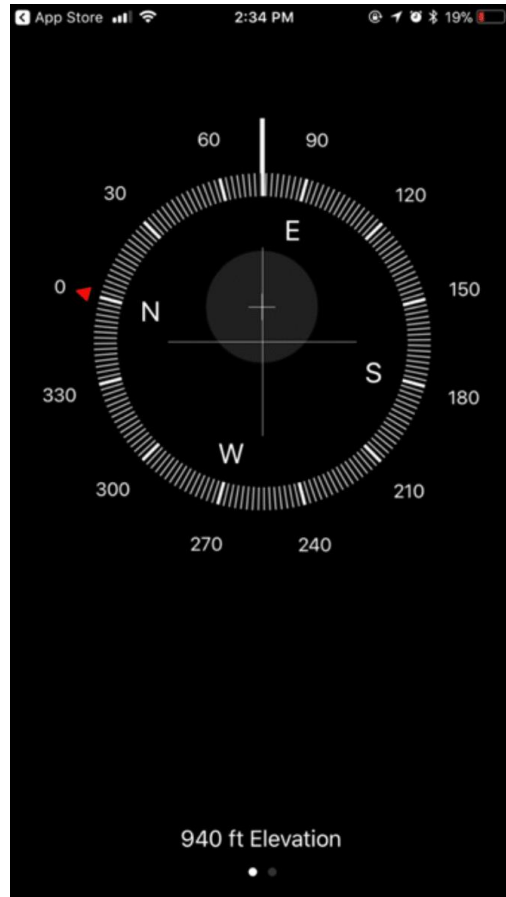
- 8 Data Centers
- 256-bit AES Data Encryption
- No Fee for After Hours Support
- Accessible Anywhere, Anytime
- 100% Renewable Data Center Power
- 99.999% Uptime Service Level Agreement
- No Fee for Client Data Export
- Up to 60 Months Price Guarantee
- SOC1/SOC2/SOC3/SSAE16 Certifications
- Unlimited Cloud Platform Support
- No Fee for Device Reconfiguration

Below the quick facts, there are four columns, each with an icon and a title:

- Law Firm Business Continuity Management**: A cloud icon with a red circle and a white 'X'. The text describes a cloud-based legal software solution offering fully integrated backup and disaster recovery cloud technology in addition to custom Business Continuity solutions that address both catastrophic and clinical events to protect your firm's confidential and irreplaceable case and client data from the unpredictable. A "Learn More" link with a right arrow is at the bottom.
- Full-Spectrum Security**: A cloud icon with a red shield. The text states that Abacus Private Cloud is complete with robust security and compliance policies, including two-factor authentication, five tiers of physical, network and data security, information handling and data privacy protocols in accordance with NIST standards, and geographically redundant, SSAE-16 and SOC-1, 2, and -3 compliant data centers. A "Learn More" link with a right arrow is at the bottom.
- Zero Capital Investment**: A cloud icon with a red 'X' and a red circle. The text explains that Private Cloud environments reduce up-front capital expenses such as server hardware and software, backup hardware and software, network infrastructure and installation fees. Leverage your operating expenses budget while speeding time to deployment with Abacus Private Cloud virtualization technology.
- Complete Ownership of Data**: A cloud icon with a red key. The text contrasts Abacus Private Cloud with other Public Cloud storage and SaaS providers that may keep you from accessing your own data, stating that with Abacus Private Cloud you own your data. Upon termination of any business relationship, you will always have access to your irreplaceable case and client data.

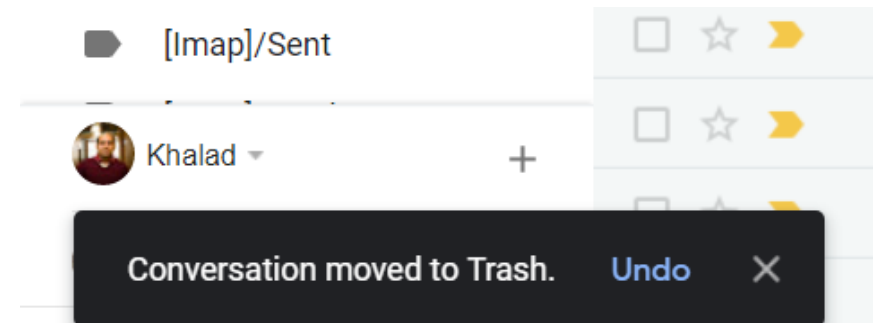
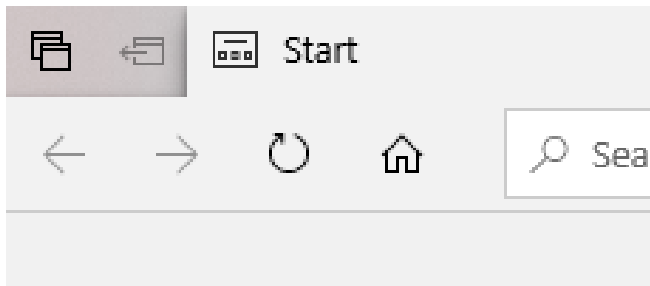
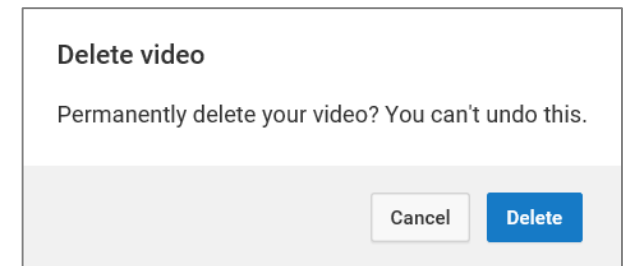
Match Between System and the Real World

- UI elements in a compass app (left) are similar to a compass in the real-world (right) and make it easy for users to understand the app's use and function.



User Control and Freedom

- “Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.”
- Used to be called “clearly marked exits”
- Webpage back and forward buttons
- Provide undo and redo
- Long operations should be cancelable
- Offer option to cancel in dialog boxes

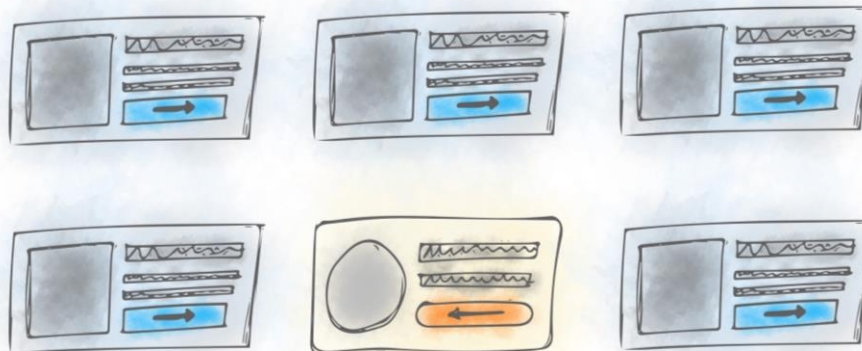


Consistency and Standards

- “Users should not have to wonder whether different words, situations, or actions mean the same thing.”
- Two key reasons for consistency and standards in UI Design:
Reduce Learning and Eliminate Confusion

Internal consistency

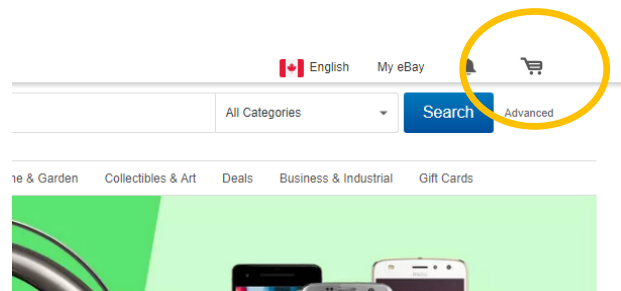
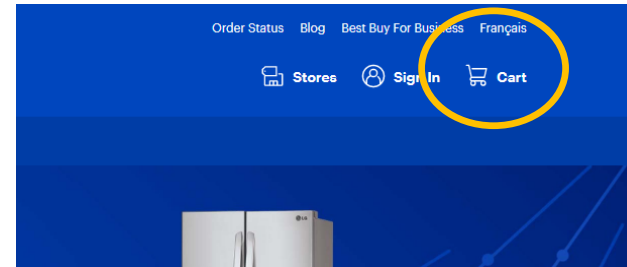
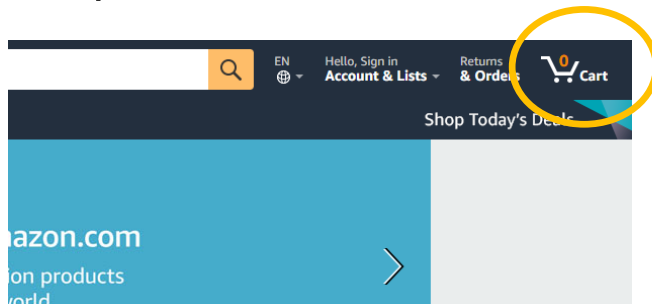
- This is the combination of both visual and functional consistency in your product design.
- It improves the usability and learnability of the product. Even when you introduce new features/pages users will have easy way using them as long as you keep the internal consistency.



Consistency and Standards

External consistency

- This type of consistency is achieved when there is design consistency across multiple systems/products.
- This way the user's knowledge for one product can be reused in another.
- This helps eliminate a lot of the friction and provides great user experience.



Error Prevention

- “Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.”



The following example is a demonstration from the usage tab

Date	99/99/9999
Phone	(123) 456-7___ (999) 999-9999
Tax ID	99-9999999

Profile

* NAME

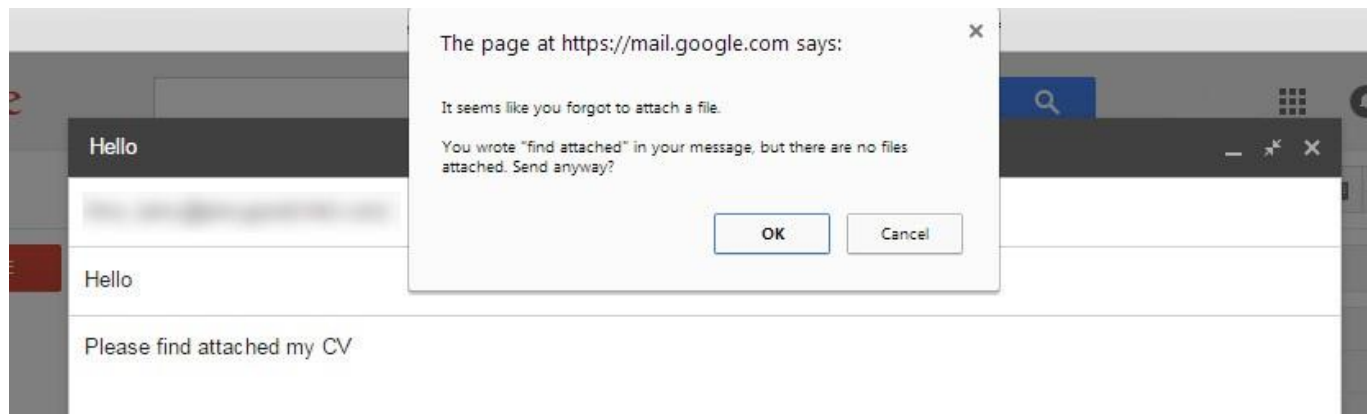
First Name Last Name

* MOBILE NUMBER

+1 (555) 666-7778

* LANGUAGE

English (United States)



Recognition Rather Than Recall

- “Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.”
- Recognition: ability to “recognize” an event or piece of information as being familiar
- Recall: retrieval of related details from memory

Your Recently Viewed Items and Featured Recommendations

You viewed

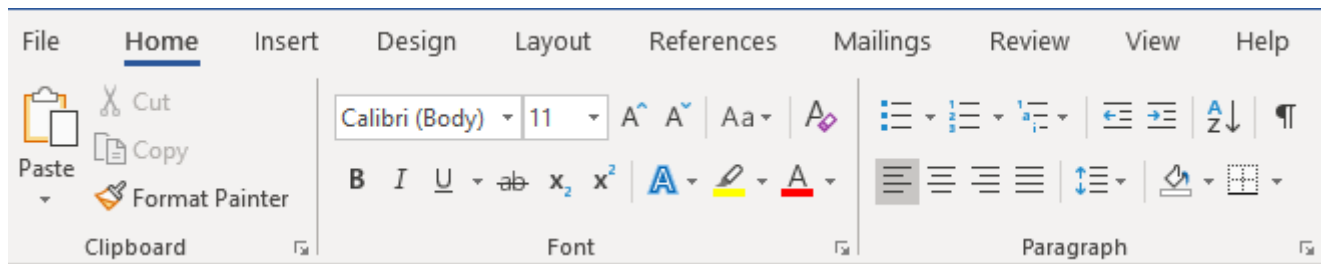


› View or edit your browsing history

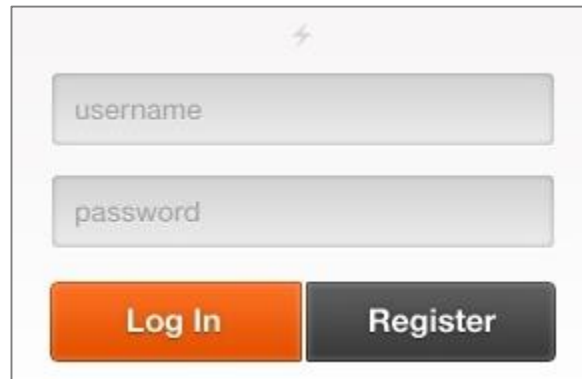
- Example: When a user goes back to Amazon.com, the personalized homepage includes a list of recently viewed items.

Recognition and Recall in User Interfaces

- A menu system is the most classic example of a recognition-based user interface: the computer shows you the available commands, and you recognize the one you want.

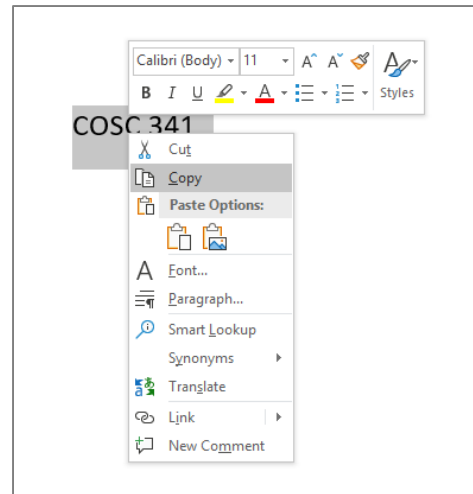
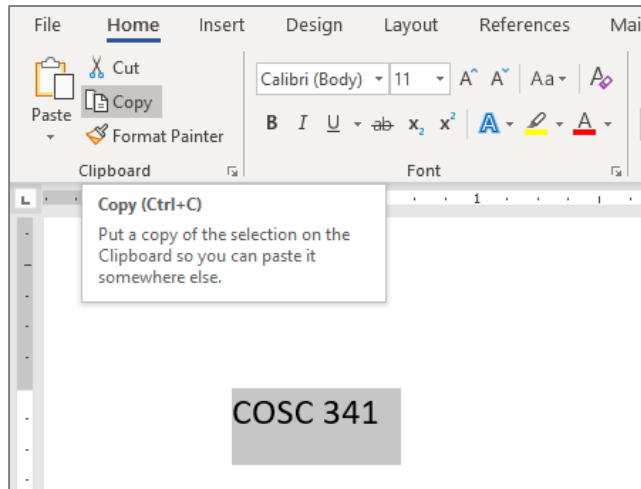


- The classic example of recall in an interface is login. When you log in to a site, you have to remember both a username (or email) and a password.

A login form with a light gray background. It features two input fields: 'username' and 'password', both with placeholder text. Below the fields are two buttons: an orange 'Log In' button and a dark gray 'Register' button. The form is enclosed in a thin gray border.

Flexibility and Efficiency of Use

- “Accelerators — unseen by the novice user — may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.”
- Flexible process can be carried out in different ways, so that people can pick whichever method fits them



Ctrl + C

Aesthetic and Minimalist Design

- “Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.”

The screenshot shows the GoDaddy homepage, which is highly cluttered. At the top, there are multiple banners and links, including a 'Reseller Plans! RELOADED' banner. Below this, there are navigation links for various services like Domain Names, Hosting & Servers, Site Builders, Email, SSL Certificates, Ecommerce, and Be a Reseller. A large banner in the center promotes 'Secure email accounts start at just \$9.95/yr!'. To the left, there is a sidebar with links to 'Domain Names', 'Web Hosting', 'Site Builders', 'Email Accounts', 'Secure SSL Certificates', 'Ecommerce', and 'Reseller Opportunities'. The bottom of the page features a footer with contact information and a 'GO' button.

The screenshot shows the GoDaddy homepage with a clean and minimalist design. The layout is organized with clear navigation links at the top, including 'All Products', 'Domains', 'Websites', 'Hosting', 'Web Security', 'Online Marketing', and 'Email & Office'. A large search bar is prominently displayed, asking 'Find your perfect domain name'. Below this, there are several promotional banners, including one for 'GET A FREE DOMAIN WHEN YOU ORDER WEBSITE BUILDER.' and another for 'WHAT WILL YOU DO WITH YOUR DOMAIN NAME? LAUNCH YOUR SITE WITH WEB HOSTING'. The design uses a lot of white space and clear typography to highlight the main offers.

Help Users Recognize, Diagnose, and Recover From Errors

- “Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.”



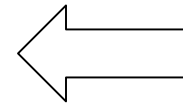
Have an Account? Login Here.

Enter login information here and click the **Login** button below.

Username:

Password:

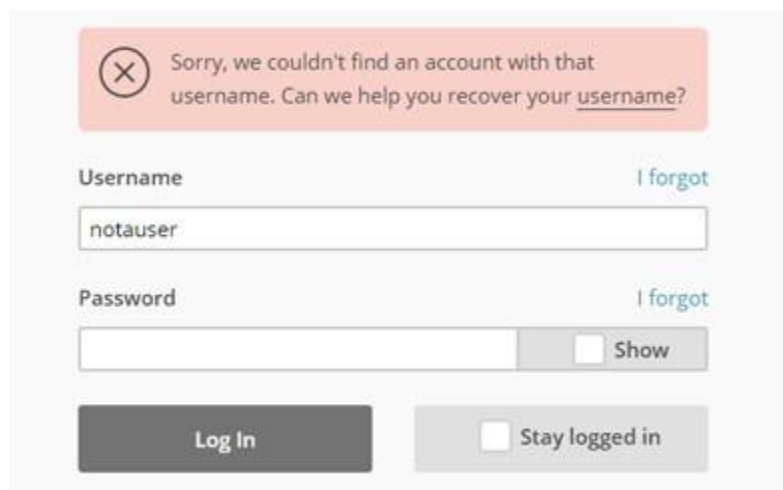
Login



what's wrong?

- username error?
- password error?

Could not login. Valid authentication credentials were not provided.

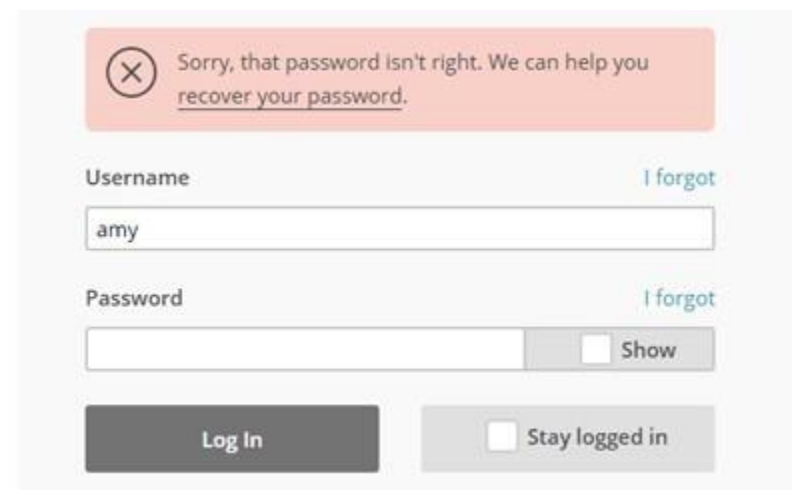


Sorry, we couldn't find an account with that username. Can we help you recover your [username](#)?

Username [I forgot](#)

Password [I forgot](#)
☐ Show

Log In ☐ Stay logged in



Sorry, that password isn't right. We can help you recover your [password](#).

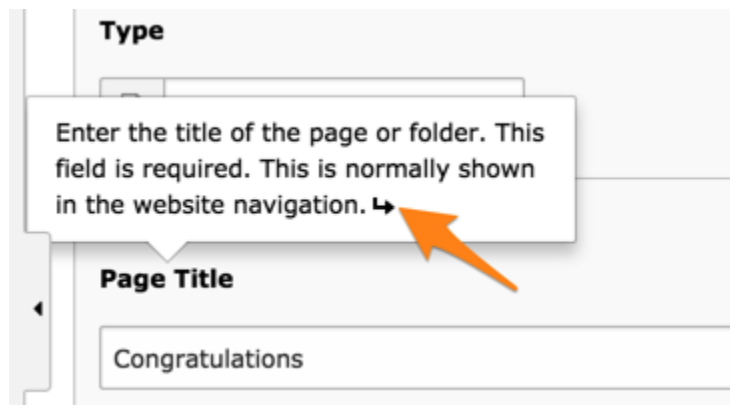
Username [I forgot](#)

Password [I forgot](#)
☐ Show

Log In ☐ Stay logged in

Help and Documentation

- “Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.”



The image shows a portion of a web form. At the top, there is a label 'Type' above a text input field. Below this, a tooltip box is displayed, containing the text: 'Enter the title of the page or folder. This field is required. This is normally shown in the website navigation. ↳'. An orange arrow points from the tooltip to the 'Page Title' label. Below the 'Page Title' label is a text input field containing the word 'Congratulations'.

Severity Rating

- Used to allocate resources to fix problems
- Estimates of need for more usability efforts
- Combination of
 - its **frequency** of occurrence (e.g., common or rare)
 - its **impact** on users (e.g., easy or hard to overcome)
 - its **persistence** (e.g., does it need to be overcome once or repeatedly)
- Should be done independently by all judges

Severity Rating

0 – don't agree that this is a usability problem

1 - cosmetic problem

2 - minor usability problem

3 - major usability problem; important to fix

4 - usability catastrophe; imperative to fix

Example: Account Information Edit

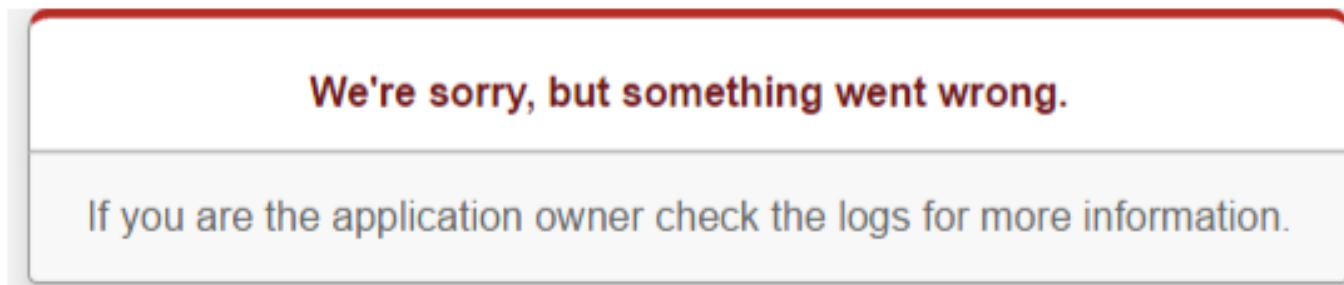
- Issue: Unable to edit year of birth
- Severity: 2 (Minor)
- Heuristics violated:
 - User control and freedom
- Description:

When you first create an account you can enter basic information about yourself. Sometimes you don't want to reveal private info so you write whatever to get pass the profile. Later, if you want to give an honest birth year, you can't find where to edit it.

Example: Website links

- Issue: Reference requesting works, but throws an error
- Severity: 3 (Major)
- Heuristics violated:
 - Error Prevention
- Description:

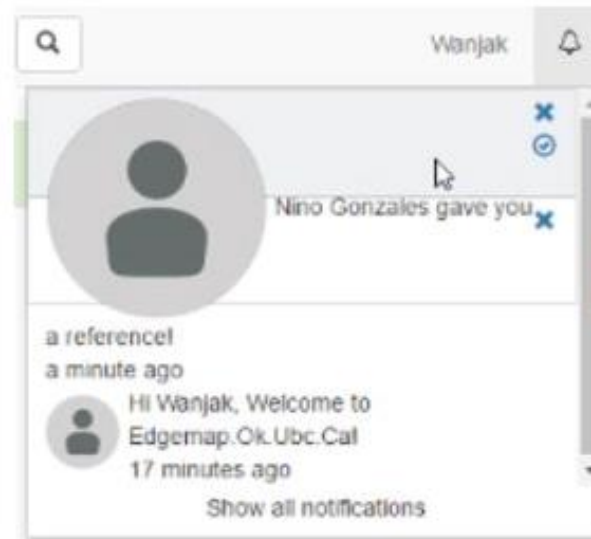
Once a user hits submit after filling out the request form, they get an error. The person they are trying to register, they'll receive the request anyway regardless of the error.



Example: Layout

- Issue: Notification Layout
- Severity: 1 (cosmetic problem)
- Heuristics violated:
 - Aesthetic and Minimalist Design
- Description:

When a user has a reference notification, they layout becomes cluttered and disorganized



Phases of Heuristic Evaluation

- Pre-evaluation training
 - give evaluators needed domain knowledge & information on the scenario
- Evaluation
 - individuals evaluates UI & makes list of problems
- Severity rating
 - determine how severe each problem is
- Aggregation
 - group meets & aggregates problems
- Debriefing
 - discuss the outcome with design team

Suggested Report Format

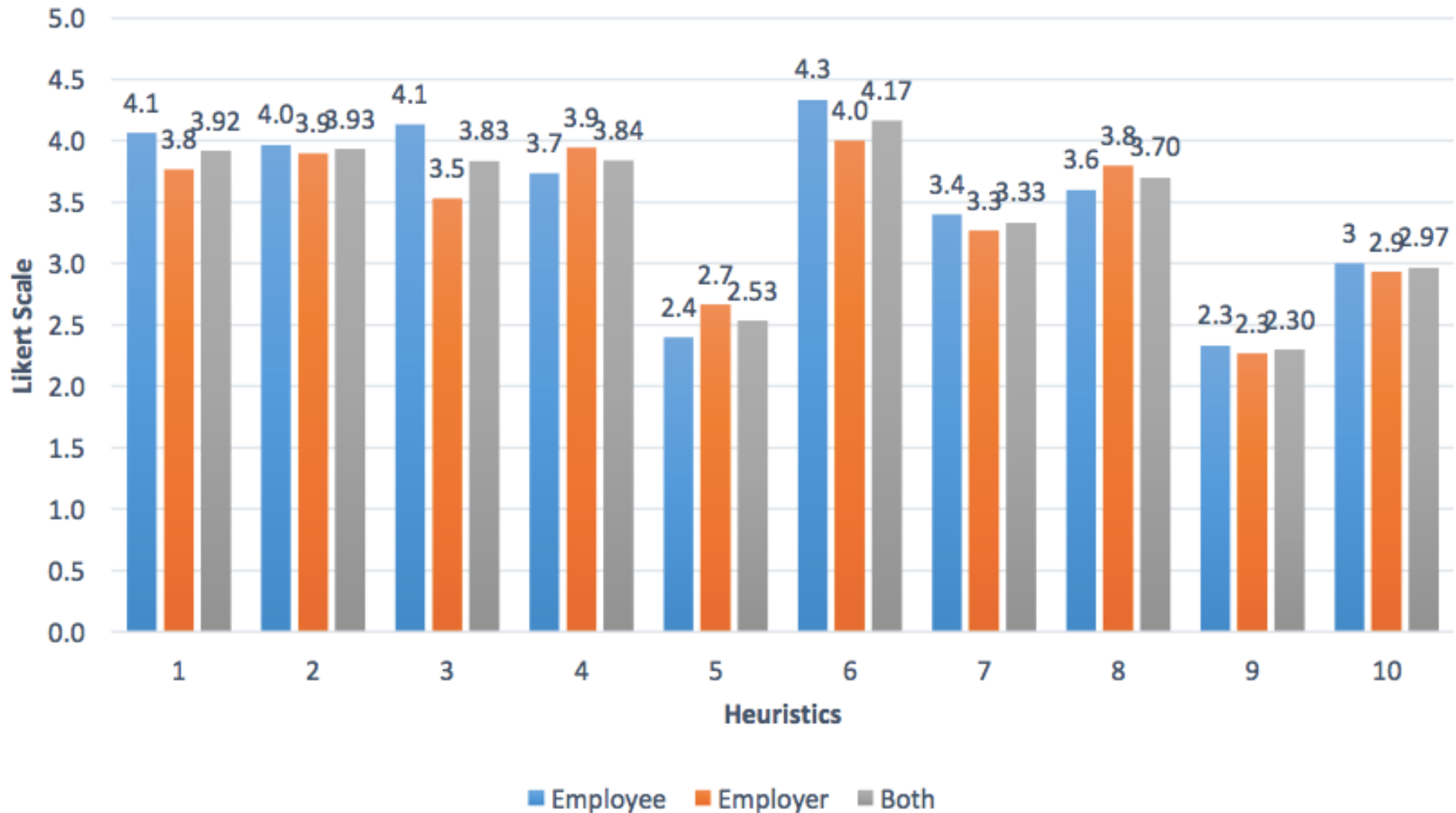
- What to include:
 - Issues/Problems
 - Heuristic
 - Severity
 - Description
 - Recommendation (if any)
 - Screenshot (if helpful)

Suggested Report Format

- Sample question:
 - I agree that this system has good design that helps users recognize the visibility of system status.
 - Responses: Strongly agree, Agree, Neutral, Disagree, Strongly disagree

Example Quantitative Results

- This evaluation had 2 user groups



Advantages and Disadvantages of Heuristics

Advantages:

- It can provide some quick and relatively inexpensive feedback to designers in the design process
- Assigning the correct heuristic can help suggest the best corrective measures to designers
- You can use it together with other usability testing methodologies

Disadvantages:

- It requires knowledge and experience to apply the heuristics effectively
- Trained usability experts are sometimes hard to find and can be expensive
- You should use multiple experts and aggregate their results

What Did We Cover

- Nielsen Heuristics
 - Visibility of System Status
 - Match Between System and the Real World
 - User Control and Freedom
 - Consistency and Standards
 - Error Prevention
 - Recognition Rather Than Recall
 - Flexibility and Efficiency of Use
 - Aesthetic and Minimalist Design
 - Help Users Recognize, Diagnose, and Recover From Errors
 - Help and Documentation
- Severity Rating
- Phases of Heuristic Evaluation
- Heuristic Report Format

Job Title

- Interaction Designer
- UX Designer
- UX Engineer
- User Experience Designer Graphics
- User Experience/Interface Designer
- User Interface Web Designer
- Architect/Researcher for Human Computer Interaction
- UX Researcher
- User Researcher
- Software Developer - Human Computer Interaction
- Product Designer
- ...

Requirements

- "Experience with mobile and tablet design"
- "Experience working with Android and iOS OS Applications"
- "Familiar with mobile platform development"
- "Design, implement, test, deploy and maintain mobile software solutions."
- "Development experience with iOS and/or Android platform"
- "In-depth understanding of mobile application development"

US

- Google (<https://research.google/research-areas/human-computer-interaction-and-visualization/>)
- Microsoft (<https://www.microsoft.com/en-us/research/group/human-computer-interaction/>)
- Facebook (<https://tech.facebook.com/reality-labs/>)
- Disney Research (<https://la.disneyresearch.com/research-area/human-computer-interaction/>)
- Autodesk (<https://www.research.autodesk.com/research-areas/research-area-science/#human>)

Companies

- Autodesk
- IBM Canada
- Huawei Canada
- EA Sports
- McAfee/Trellix
- Amazon
- Shopify
- Mobify
- Ubisoft
- Thalmic Labs Inc
- Mastercard
- Element AI

Job title (Mobile)

- Mobile Developer
- Mobile UX Architect
- Mobile Apps Developer
- Android Developer
- Mobile Application Developer
- Mobile Application Developer(Architect)
- Application Developer-Mobile for iOS and Android
- Mobile Application Developer - Android

Question?